

Multivariable Calculus

Partial Derivatives & Chain Rule

Course: Multivariable Calculus | **For:** Computer Science - 3rd Semester

Topics Covered: Partial Differential Equations (First, Second & Third Derivatives), Mixed Partial Derivatives, Trigonometric Functions, Scenario-Based Applications, and the Multivariable Chain Rule.

Section	Topic
1	Partial Derivatives - Basics (Single & Multivariable)
2	Geometrical Interpretation
3	Higher-Order Partial Derivatives (2nd & 3rd)
4	Partial Derivatives of Trigonometric Functions
5	Mixed Derivatives & Independence/Dependence of Variables
6	Practice Questions
7	Scenario-Based / Case Study Problems
8	The Multivariable Chain Rule
9	Chain Rule - Worked Examples

Week 2 – Lecture 1: Partial Differential Equation (Introduction)

1.1 First Derivative – Single Variable

For an ordinary function of one variable, the derivative is found in the usual way.

$$f(x) = 2x + 1$$

$$df/dx = 2x + 1 = 2 \, dx/dx + 0$$

$$df/dx = 2$$

1.2 First Derivative – Multivariable (Partial Derivative)

When a function depends on more than one variable, we use **partial derivatives** to find the derivative with respect to one variable while treating all other variables as constants.

$$f(x, y) = 2x + 3y$$

$$\text{w.r.t 'x': } \partial f / \partial x = f_x = 2(1) + \text{Constant} = 2 + 0 = 2$$

$$\text{w.r.t 'y': } \partial f / \partial y = f_y = \text{Constant} + 3(1) = 0 + 3 = 3$$

Note: For multivariable derivatives, we use partial derivatives to find the derivative of a function with respect to one of its variables, while keeping the other variables constant.

1.3 Geometrical Interpretation

Geometrically, a partial derivative represents the slope of the surface $z = f(x, y)$ in the direction of one axis (x or y) at a fixed point, while moving parallel to that axis and holding the other coordinate constant.

1.4 Worked Example

Question 2: $f(x, y) = x^3 + y^3$

$$f_x = 3x^2 + 0 = 3x^2 \quad (\text{w.r.t 'x'})$$

$$f_y = 3y^2 + 0 = 3y^2 \quad (\text{w.r.t 'y'})$$

1.5 Partial Derivatives – Three Variables

$$f(x, y, z) = x^2 + y^2 + z^2$$

$$f_x = 2x \quad (\text{w.r.t } x) \quad f_y = 2y \quad (\text{w.r.t } y) \quad f_z = 2z \quad (\text{w.r.t } z)$$

1.6 Second Derivative

$$\partial f / \partial x = \partial^2 f / \partial x \partial x = f_{xx}$$

From $f_x = 2x \rightarrow f_{xx} = 2$

From $f_y = 2y \rightarrow f_{yy} = 2$

1.7 Third Derivative – Practice Questions

Q1. $f(x, y, z) = x^2y + yz^3$. Find $f_{xxx}, f_{yyy}, f_{zzz}$.

w.r.t x	w.r.t y	w.r.t z
$f_x = 2xy$	$f_y = x^2 + z^3$	$f_z = 3z^2y$
$f_{xx} = 2y$	$f_{yy} = 0$	$f_{zz} = 6zy$
$f_{xxx} = 0$	$f_{yyy} = 0$	$f_{zzz} = 6y$

1.8 Partial Derivatives of Trigonometric Functions

Q1. $f(x, y, z) = \sin(xyz)$. Find $f_{xxx}, f_{yyy}, f_{zzz}$.

Differentiate w.r.t 'x':

$$f_x = \cos(xyz) \cdot yz = yz \cdot \cos(xyz)$$

$$f_{xx} = yz[-\sin(xyz)] \cdot yz = -y^2z^2\sin(xyz)$$

$$f_{xxx} = -y^2z^2\cos(xyz) \cdot yz = -y^3z^3\cos(xyz)$$

Differentiate w.r.t 'y':

$$f_y = xz \cdot \cos(xyz)$$

$$f_{yy} = xz[-\sin(xyz)] \cdot xz = -x^2z^2\sin(xyz)$$

$$f_{yyy} = -x^3z^3\cos(xyz)$$

Differentiate w.r.t 'z':

$$f_z = xy \cdot \cos(xyz)$$

$$f_{zz} = x^2y^2[-\sin(xyz)]$$

$$f_{zzz} = -x^3y^3\cos(xyz)$$

Week 2 – Lecture 2: Higher-Order Derivatives & Mixed Derivatives

2.1 Worked Example

Q. $f(x, y, z) = x \cdot \sin(2y + 3z)$. Find $f_{xxx}, f_{yyy}, f_{zzz}$.

w.r.t 'x': $f_x = \sin(2y+3z), f_{xx} = 0, f_{xxx} = 0$

w.r.t 'y': $f_y = 2x \cdot \cos(2y+3z)$

$$f_{yy} = -4x \cdot \sin(2y+3z)$$

$$f_{yyy} = -8x \cdot \cos(2y+3z)$$

w.r.t 'z': $f_z = 3x \cdot \cos(2y+3z)$

$$f_{zz} = -9x \cdot \sin(2y+3z)$$

$$f_{zzz} = -18x \cdot \cos(2y+3z)$$

Note: 1. Concave up (■) tells us **minima**. 2. Concave down (■) tells us **maxima**. 3. For maximum/minimum values, we find the **second derivative**.

2.2 Mixed Derivatives

$$f(x, y) = x^2 + y^2$$

$$f_x = 2x, \quad f_y = 2y$$

$$f_{xy} = 0 \quad (\text{this is the mixed derivative})$$

2.3 Interpreting the Mixed Derivative f_{xy}

$f(x, y) = x^2 + y^2$ can be thought of as a model where x and y are independent risk factors.

- If $f_{xy} = 0 \rightarrow$ the function is **separable**. A mixed derivative of zero means there is no interaction between x and y — in simple terms, the risk factors are **independent**.
- If $f_{xy} \neq 0 \rightarrow$ the function is **not separable**, indicating an interaction between x and y — in simple terms, the risk factors are **dependent**.

2.4 Example – Server Response Time

$$f(x, y) = x^3 + y^3 \quad (x = \text{User}, y = \text{Data})$$

$$f_x = 3x^2, \quad f_y = 3y^2, \quad f_{xy} = 0$$

$\Rightarrow$ User and data are **independent**.

2.5 Practice Questions – Two & Three Variables

Q1 (Two variables). $f(x, y) = x^2y$. Find f_x, f_y, f_{xy} .

$$f_x = 2xy, \quad f_y = x^2, \quad f_{xy} = 2x \rightarrow \text{Dependent}$$

Q2 (Three variables). $f(x, y, z) = x^2y + yz^3$. Find $f_x, f_{xx}, f_y, f_{yy}, f_{xyz}$.

$$f_x = 2xy, \quad f_{xx} = 2y, \quad f_y = x^2 + z^3, \quad f_{yy} = 0$$

$$f_x = 2xy \quad \& \quad f_{yx} = 2x, \quad f_{xyz} = 0 \rightarrow \text{Independent}$$

Q3. $f(x, y) = x^2y + 3xy^2$. Find $f_x, f_{xx}, f_y, f_{yy}, f_{xy}, f_{yx}$.

Derivative	Result
f_x	$2xy + 3y^2$
f_{xx}	$2y$
f_y	$x^2 + 6xy$
f_{yy}	$6x$
f_{xy}	$2x + 6y \quad (\text{Dependent})$
f_{yx}	$2x + 6y \quad (\text{Dependent})$

Scenario-Based / Case-Study Problems

These applied problems connect partial derivatives to real Computer-Science and AI/ML contexts.

3.1 AI Model Training Cost

Problem: An AI company models its training cost as a function of dataset size (x) and number of layers (y): $f(x, y) = 3x^2y + 2y^3$.

Task: Find f_x, f_y, f_{xy} , and interpret f_{xy} .

$$f_x = 6xy, \quad f_y = 3x^2 + 6y^2, \quad f_{xy} = 6x$$

Interpretation: As dataset size increases, each additional layer adds more cost. This shows that dataset size and model architecture complexity **interact** (dataset and layers are dependent).

3.2 Server Response Time Model

Problem: At TechNova, server response time depends on CPU usage (x) and memory load (y): $f(x, y) = x^2 + y^2$.

Task: Find f_x , f_y , and interpret the result.

$$f_x = 2x, \quad f_y = 2y$$

Interpretation: Response time increases linearly with CPU and memory load. If CPU or memory load doubles, response time quadruples.

3.3 Cloud Resource Utilization

Problem: A cloud engineer defines total system load as a function of CPU, memory, and network latency: $f(x, y, z) = x^3 + y^3 + z^3$.

Task: Find f_x , f_y , f_z , and f_{xyz} .

$f_x = 3x^2$	$f_y = 3y^2$	$f_z = 3z^2$	$f_{xyz} = 0$
$f_{xx} = 6x$	$f_{yy} = 6y$	$f_{zz} = 6z$	$f_{xyz} = 0$
$f_{xxx} = 6$	$f_{yyy} = 6$	$f_{zzz} = 6$	$f_{xyz} = 0$

Interpretation: Each factor affects system load independently. No interaction exists between CPU, memory, and network.

3.4 AI Inference Latency

Problem: The inference latency (in ms) of an AI model is modeled as: $f(x, y, z) = x^2y + yz^3$.

Task: Find f_x , f_y , f_z , f_{xyz} , and interpret f_{xyz} .

$$f_x = 2xy, \quad f_y = x^2 + z^3, \quad f_z = 3yz^2$$

$$f_{xy} = 2x, \quad f_{xyz} = 0$$

Interpretation: CPU, memory, and network affect latency separately. If $f_{xyz} \neq 0$, it would indicate interdependence among these variables.

Week 3 – Lecture 1: The Multivariable Chain Rule

4.1 Formula

Let $w = f(x, y)$ where $x = x(t)$ and $y = y(t)$. Then the total derivative of f with respect to t is:

$$df/dt = (\partial f/\partial x)(dx/dt) + (\partial f/\partial y)(dy/dt)$$

- ∂ (partial) means a **partial** derivative — one variable at a time, others held constant.
- d (as in df/dt) means the **total** derivative — the overall rate of change with respect to t .

The relationship can be visualized as a tree diagram: $f(x, y)$ branches into $\partial f/\partial x$ and $\partial f/\partial y$, and each of x and y branches down to t via dx/dt and dy/dt .

4.2 Worked Examples

Example (i). $f(x, y) = x^2 + y^2$, where $x(t) = t^2$, $y(t) = t$. Find df/dt .

$$\partial f/\partial x = 2x, \quad \partial f/\partial y = 2y, \quad dx/dt = 2t, \quad dy/dt = 1$$

$$df/dt = 2x \cdot 2t + 2y \cdot 1$$

Substitute $x = t^2$, $y = t$:

$$df/dt = 2(t^2) \cdot 2t + 2(t) = 4t^3 + 2t$$

$$df/dt = 4t^3 + 2t$$

Example (ii). $f(x, y) = x^2y + y^3$, where $x(t) = t^2$, $y(t) = 3t$. Find df/dt .

$$\partial f/\partial x = 2xy, \quad \partial f/\partial y = x^2 + 3y^2, \quad dx/dt = 2t, \quad dy/dt = 3$$

$$df/dt = 2xy \cdot 2t + (x^2 + 3y^2) \cdot 3$$

Substitute $x = t^2$, $y = 3t$:

$$= [2(t^2)(3t)] \cdot 2t + [(t^2)^2 + 3(3t)^2] \cdot 3$$

$$= 6t^3 \cdot 2t + (t^4 + 27t^2) \cdot 3 = 12t^4 + 3t^4 + 81t^2$$

$$\mathbf{df/dt = 15t^4 + 81t^2}$$

Example (iii). $f(x, y) = x^3 + y^3$, where $x(t) = t^3$, $y(t) = 2t$. Find df/dt .

$$\partial f/\partial x = 3x^2, \quad \partial f/\partial y = 3y^2, \quad \partial x/\partial t = 3t^2, \quad \partial y/\partial t = 2$$

$$df/dt = 3x^2 \cdot 3t^2 + 3y^2 \cdot 2 = 3(t^3)^2 \cdot 3t^2 + 3(2t)^2 \cdot 2$$

$$\mathbf{df/dt = 9t^8 + 24t^2}$$

Example (iv). $f(x, y) = 2xy^2 + 3x^2y$, where $x(t) = 2t^5$, $y(t) = 5t^2$. Find df/dt .

$$\partial f/\partial x = 2y^2 + 6xy, \quad \partial f/\partial y = 4xy + 3x^2, \quad \partial x/\partial t = 10t^4, \quad \partial y/\partial t = 10t$$

$$df/dt = (2y^2 + 6xy)(10t^4) + (4xy + 3x^2)(10t)$$

Substituting $x = 2t^5$, $y = 5t^2$ and simplifying:

$$\mathbf{df/dt = 900t^8 + 720t^{11}}$$

Example (v). $f(x, y) = \sin(xy)$, where $x(t) = t^2$, $y(t) = t^3$. Find df/dt .

$$\partial f/\partial x = y \cdot \cos(xy), \quad \partial f/\partial y = x \cdot \cos(xy), \quad \partial x/\partial t = 2t, \quad \partial y/\partial t = 3t^2$$

$$df/dt = y \cdot \cos(xy) \cdot 2t + x \cdot \cos(xy) \cdot 3t^2$$

$$= t^3 \cdot \cos(t^5) \cdot 2t + t^2 \cdot \cos(t^5) \cdot 3t^2 = 2t^4 \cos(t^5) + 3t^4 \cos(t^5)$$

$$\mathbf{df/dt = 5t^4 \cdot \cos(t^5)}$$

Example (vi). $f(x, y) = \sin(x^3y^3)$, where $x(t) = t^2$, $y(t) = t^3$. Find df/dt .

$$\partial f/\partial x = \cos(x^3y^3) \cdot 3x^2y^3, \quad \partial f/\partial y = \cos(x^3y^3) \cdot 3x^3y^2$$

$$\partial x/\partial t = 2t, \quad \partial y/\partial t = 3t^2$$

$$df/dt = [\cos(x^3y^3) \cdot 3x^2y^3] \cdot 2t + [\cos(x^3y^3) \cdot 3x^3y^2] \cdot 3t^2$$

Substituting $x = t^2$, $y = t^3$ gives $x^3y^3 = t^{15}$:

$$= 6t^{14} \cos(t^{15}) + 9t^{14} \cos(t^{15})$$

$$\mathbf{df/dt = 15t^{14} \cdot \cos(t^{15})}$$